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ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 4 SET A **WINTER SEMESTER, 2024–2025**

DURATION: 30 MINUTES **FULL MARKS: 20**

Math 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. Find the Singular Value Decomposition (SVD) ($A = U\Sigma V^T$) for $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ 10 (CO3) (PO1)

Solution:

Given, $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$

So, $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$.

Therefore, $AA^T = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ and $A^TA = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

For AA^T, Let the eigenvalues be λ_1, λ_2 and λ_3. Thus,

$$\lambda^3 - (trace(A)\lambda^2) + (\sum PrincipalMinors)\lambda - det(A) = 0$$

$$\Rightarrow \lambda^3 - 4\lambda^2 + (1+1+1)\lambda - 0 = 0$$

Therefore, $\lambda_1 = 3, \lambda_2 = 1$ and $\lambda_3 = 0$ (solving the equation and in decreasing order)

For $\lambda_1 = 3$,

$$AA^T - 3I = \begin{bmatrix} -1 & -1 & 1 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -1 & 1 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Setting $x = 1$ and solving first two equations, $y = -1/2$ and $z = -1/2$. Too keep things simpler, we multiply all values by 2. So, $u_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$.

Normalized $u_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$.

For $\lambda_1 = 1$,

$$AA^T - I = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 1 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solution:

This is more intuitive, all 0s in 2nd and 3rd entry of 2nd and 3rd row, so to keep it consistent, right side must be 0 as well. So, $x = 0$ and y and z must be same, so 1. (other values are also acceptable)

So, $u_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

Normalized $u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

For $\lambda_3 = 0$,

$$AA^T - 0I = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Setting $x = 1$ and solving other values, So, $u_3 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$.

Normalized $u_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$.

So, $U = \begin{bmatrix} 2/\sqrt{6} & 0 & 1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \end{bmatrix}$

Now, $A^T A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

For $\lambda_1 = 3$,

$$A^T A - 3I = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix}$$

$$A^T A - 3I = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

So, $v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. (intuitively)

Normalized $v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

For $\lambda_2 = 1$,

$$A^T A - I = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$A^T A - 3I = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

So, $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. (intuitively)

Normalized $v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

So, $V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

And $\Sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

So,

$$A = U\Sigma V^T$$

$$\Rightarrow A = \begin{bmatrix} 2/\sqrt{6} & 0 & 1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & -1/\sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}^T$$

Rubrics:

- 1 point each (times 3) for correctly identifying each eigenvalue.
- 1 point each (times 7)(2+2+3) for correctly determining each eigenvector.

2. A sequence is given by 10 (CO3) (PO1)

$$1,\ 5,\ 7,\ -1,\ -23, \cdots$$

where each term is obtained as a linear combination of the two preceding terms. The first two terms of the sequence are $a_0 = 1$ and $a_1 = 5$.

a) Determine the recurrence relation for the sequence. In other words, express a_n in terms of a_{n-1} and a_{n-2}.

b) Express the recurrence relation obtained in Question 2.a in matrix form and use diagonalization to derive a formula for the k-th term of the sequence.

c) Using the result obtained in Question 2.b, compute the 10th term of the sequence and verify that $a_{10} = -329$.

Solution:

Let the coefficients of the two terms be x and y, So, $xa_{n-2} + ya_{n-1} = a_n$ For the first two terms, $n = 2$ and $n = 3$

$$x + 5y = 7 \tag{1}$$

$$5x + 7y = -1 \tag{2}$$

Solving the two equations we get, $x = -3$ and $y = 2$.

$$F_{n+1} = -3F_{n-1} + 2F_n$$

$$F_n = F_n$$

So,

$$\begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ 1 & 0 \end{bmatrix}^n \begin{bmatrix} F_1 \\ F_0 \end{bmatrix}$$

Let $A = \begin{bmatrix} 2 & -3 \\ 1 & 0 \end{bmatrix}$ So, $A^n = \begin{bmatrix} 2 & -3 \\ 1 & 0 \end{bmatrix}^n$

And, $A^n = S\Lambda^n S^{-1}$

We know, $\lambda^2 - trace(A)\lambda + det(A) = 0$

so, $\lambda^2 - 2\lambda + 3 = 0$

Solving, we get, $\lambda_1 = 1 - \sqrt{2}i$ and $\lambda_2 = 1 + \sqrt{2}i$

$A - \lambda I = \begin{bmatrix} 2-\lambda & -3 \\ 1 & -\lambda \end{bmatrix}$

And $\begin{bmatrix} 2-\lambda & -3 \\ 1 & -\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

We can see that, if $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \lambda \\ 1 \end{bmatrix}$ it is satisfied.

And $\begin{bmatrix} 2-\lambda & -3 \\ 1 & -\lambda \end{bmatrix} \begin{bmatrix} \lambda \\ 1 \end{bmatrix} = \begin{bmatrix} 2\lambda - \lambda^2 - 3 \\ \lambda - \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

So, the eigenvectors for eigenvalues λ_1 and λ_2 are $\begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix}$ respectively.

So, $S = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix}$.

Therefore, $A = S\Lambda S^{-1}$

or, $A^n = S\Lambda^n S^{-1}$

Now,

$$\begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = A^n \begin{bmatrix} F_1 \\ F_0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = S\Lambda^n S^{-1} \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 1 & \lambda_2 \end{bmatrix}^n \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 1 & \lambda_2 \end{bmatrix}^n \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^n & 0 \\ 1 & \lambda_2^n \end{bmatrix} \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^n & 0 \\ 1 & \lambda_2^n \end{bmatrix} \begin{bmatrix} 5 - \lambda_2 \\ -5 + \lambda_1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1^n(4-\lambda_2) \\ -\lambda_2^n(4-\lambda_1) \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} \lambda_1^{n+1}(4-\lambda_2) - \lambda_2^{n+1}(4-\lambda_1) \\ \lambda_1^n(5-\lambda_2) - \lambda_2^n(5-\lambda_1) \end{bmatrix}$$

$$\boxed{F_n = \frac{1}{\lambda_1 - \lambda_2}[\lambda_1^n(5-\lambda_2) - \lambda_2^n(5-\lambda_1)]}$$

$\lambda_1 = 1 - \sqrt{2}i$ and $\lambda_2 = 1 + \sqrt{2}i$
So, $\lambda_1 - \lambda_2 = -2\sqrt{2}i$
$5 - \lambda_1 = 4 + \sqrt{2}i$ and $5 - \lambda_2 = 4 - \sqrt{2}i$
Therefore, $\mathrm{F}_n = \frac{1}{2\sqrt{2}i}[(1-\sqrt{2}i)^n(4-\sqrt{2}i) - (1+\sqrt{2}i)^n(4+\sqrt{2}i)$
For $n = 10$, you will get $F_n = -329$.

Rubrics:

- 2 points for correctly identifying the recurrence relation.
- 2 points for correctly identifying eigenvalues.
- 5 points for correctly finding out the $n-th$ or $k-th$ term of the sequence.
- 1 point for verification.